Graphic Era HILL UNIVERSITY
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University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B. Tech ECE 3rd Semester
Name of the Paper: Digital System Design
Paper Code: TEC 302

Time: 3 hour | Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. CO1

i. $(928.75)_{10} = (?)_8$
ii. $(CDBA.B7C5)_{16} = (?)_8$
iii. $(10111.001)_2 = (?)_{10}$
iv. $(1011011)_{gray} = (\quad)_2$
v. $(AEC)_{16} = (\quad)_8$

b. Explain C01
(i) Weighted and non-weighted codes
(ii) Even and Odd Parity

c. Draw all basic gates using NAND and NOR gates. Write Boolean laws. CO1

Q2. (2X10=20 Marks)

a. Use Tabular method and find the minimal expression CO2
$F(A.B.C.D) = \Sigma 1.2.3.5.9.10.12.14 + \Sigma \emptyset 0,4,6,8,11$

b. Explain 3 Variable K Map. CO2
Use K map and find the minimal expression
$F(A.B.C.D) = \Sigma\ 0.1.2.3,5,10,12,13 + \Sigma \emptyset\ 4,6,11,14$

c. CO2
(i) Realise the function using only NAND gate.
$F(X, Y, Z) = (X'+Y')(X'+Z)(Y'+Z)$
(ii) Reduce to minimal value F (A, B,C,D) $A'C'+ABD+BC'D+AB'D'+ABCD'$

Q3. (2X10=20 Marks)

a. Explain the working of 1 to 4 demultiplexer. Design full adder using 1 to 8 demultiplexer. C03

b. Design BCD to Excess-3 code converter. CO3

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c. CO3

(i) Design a combinational circuit that takes three inputs A, B, C and produces a single output F. The output F is 1 if the binary number represented by the inputs is a prime number (2, 3, 5, 7); otherwise F is 0.

(ii) Explain the working of 2 to 4 decoder.

Q4. (2X10=20 Marks)

a. Explain the working of JK and T flip flop. Draw the excitation table of SR and D flip flop. C04

b. Explain C04

Serial In Serial Out shift register.

Parallel In Parallel Out shift register.

c. Design and explain the working of 3-bit asynchronous Down counter. C04

Q5. (2X10=20 Marks)

a. CO5

(i) Design Inverter (NOT gate) using CMOS technology.

(ii) Design 2 input OR gate using CMOS technology.

b. Explain CMOS Technology used for the designing of logic gates with the help of example. C05

c. Explain the working of Random Access Memory with the help of circuit diagram. C05